

11-Requirement Compliance & Gap Audit

Resilient & Trustworthy Online Assessment Ecosystem

Audit Date:	October 07, 2026	Overall Compliance:	75.0% (Strong Partial)
Target System:	ExamResQ Prototype v2.4	Final Verdict:	ALMOST READY
Audit Framework:	11 Ecosystem Mandates	Audited Requirements:	11 of 11 Examined
Auditor Roles:	Product Auditor, Solution Architect, QA & Evaluation Panel	Status Distribution:	3 FULL / 8 PARTIAL / 0 MISSING

EXECUTIVE MANDATE & CONTEXT

This independent compliance audit rigorously evaluates the current ExamResQ application against the 11 core requirements mandated for high-stakes, resilient digital assessments. The audit inspects concrete source code, local state machines, client cryptographic operations, computer-vision proctoring pipelines, mesh networking services, and operator decision consoles. Features represented purely through static data or decorative UI are unvarnished and classified as partial demonstrations rather than production completions. This report establishes the exact implementation baseline and provides an actionable engineering roadmap to achieve 100% compliance.

3 FULL COMPLIANCE Fully Demonstrated	8 PARTIAL COMPLIANCE Functional / Mock Split	0 UNADDRESSED Completely Missing	75.0% WEIGHTED SCORE High Competitive Tier
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1. Executive Summary

ExamResQ presents an exceptionally strong, well-architected solution prototype tailored for hackathon judging and educational board procurement. Unlike traditional prototypes that rely strictly on static mockups, ExamResQ incorporates real, operational frontend capabilities: real-time in-browser TensorFlow object detection (COCO-SSD), real-time Web Audio API acoustic leak analysis, peer-to-peer WebRTC / BroadcastChannel multi-terminal mesh sync, responsive offline-safe state management, and complete incident lifecycle decision consoles.

However, when audited under strict institutional compliance rules (where features cannot be presumed complete solely from UI representations or simulated client datasets), 8 out of 11 requirements fall into the **PARTIAL** category. The primary compliance bottlenecks stem from the absence of a live persistent cloud backend database (PostgreSQL/Redis), simulated rather than cryptographic Merkle-tree recalculations on actual storage bytes, and hardcoded early-warning risk prediction arrays.

Core Strengths:	• Active in-browser AI computer vision & audio proctoring without external server dependency. • Zero-panic candidate communication during outages (timer freeze, offline buffer notification). • Real multi-terminal mesh discovery across tabs and laptops with live camera frames. • Exceptional human-centric governance (invigilator review queues rather than blind AI disqualifications).
Key Weaknesses:	• Early prediction probabilities (84%, 62%) are hardcoded mock arrays rather than rolling ML models. • Merkle root ledger & IndexedDB encryption are demonstrated through client state rather than hardened cryptographic storage. • Reconciliation engine runs on memory objects rather than reconciling real local storage deltas against server logs. • Cross-centre aggregate telemetry (38 test centres) is static mock data.
Top 5 Gap Priorities:	1. (P0) Connect live rolling time-series calculations to the Early Detection Risk Dashboard. 2. (P0) Implement genuine WebCrypto SHA-256 Merkle root hashing on the candidate answer dictionary. 3. (P1) Wire IndexedDB persistence to survive full browser/tab process crashes. 4. (P1) Link incident escalation webhooks to real external dispatch endpoints (e.g. simulated server socket). 5. (P1) Export downloadable official PDF audit certificates directly from the Audit Trust explorer.

2. Requirement Compliance Matrix

#	Requirement Description	Status	Score	Evidence Module	Primary Gap	P
01	Real-Time Monitoring	PARTIAL	75%	LiveCandidateMonitor, multiCandidateMeshService	Cross-centre stats (38 centres) are static mock objects.	P1
02	Early Detection & Prediction	PARTIAL	50%	EarlyDetectionDashboard, initialPredictiveCentres	Prediction scores (84%, 62%) are hardcoded, not computed.	P0
03	Automated Incident Escalation	PARTIAL	75%	IncidentCenter, ResilienceContext	Incidents lifecycle simulated via state, no real external webhooks.	P1
04	Backup & Disaster Recovery	PARTIAL	75%	RecoveryCenter, ResponseProtectionWidget	Client sandbox uses memory/localStorage instead of hardened IndexedDB.	P1
05	Secure Tamper-Evident Storage	PARTIAL	50%	AuditTrust, sampleAuditTrail	Tamper detection button is a scripted event, not crypto check on DB.	P0
06	Intelligent Suspicious Patterns	FULL	100%	AIProctoringHUD, SuspiciousPatternCenter	Fully demonstrable with live TF COCO-SSD & human review queue.	P2

07	Automated Reconciliation	PARTIAL	50%	ReconciliationCenter, sampleReconciliationRecords	Reconciliation engine runs on mock state array rather than DB deltas.	P1
08	Candidate Communication	FULL	100%	LiveExam, NotificationToast, StudentAssistanceModal	Complete dynamic two-way communication and non-alarming UI.	P3
09	Rescheduling Decision Support	PARTIAL	75%	DecisionSupportCenter, sampleDecisionSupportRecords	Action sign-off exists; recommendations derived from mock incidents.	P1
10	Fairness & Consistency Parity	FULL	100%	ResilienceContext, LiveExam, DecisionSupportCenter	Automated timer freeze, parity credit, and grievance resolution active.	P2
11	Post-Exam Audit & Reporting	PARTIAL	75%	Reports, AuditTrust, SubmissionReceiptModal	Complete on-screen audit ledger and receipt; lacks native PDF export.	P1

3. Detailed Requirement Audit

REQUIREMENT 01: REAL-TIME MONITORING	PARTIALLY COMPLIANT (Strong Partial) (75%)	Priority: P1 — High
Current Implementation:	ExamResQ provides operational visibility across active student workstations. In LiveCandidateMonitor, administrators view candidate statuses (■ Safe vs ■ Review), roll numbers, station IDs, answering progress, and live video feeds. The custom multiCandidateMeshService uses browser BroadcastChannel and WebRTC PeerJS to automatically discover candidate tabs on the same computer or across networked laptops.	Verified In: Verified LiveCandidateMonitor, multiCandidateMeshService, CCTV Wall mode, and Stations table. Verified Outage Simulation switches station to 'Offline Buffered' in real-time.
Identified Gap:	The wider network view in CentreMonitoring and OperationsDashboard presents hardcoded data for 38 regional centres rather than live gateway pings. True discovery across non-local network subnets requires an active WebRTC turn/signaling server.	Impact: Judges testing multi-tab local scenarios see genuine real-time capabilities, but enterprise evaluators will detect that regional centre metrics are pre-baked.
Actionable Recommendation:	Maintain the local mesh for demo reliability, but implement a lightweight simulated WebSocket telemetry stream generating jitter and latency fluctuations for the 38 background test centres.	Next Sprint
REQUIREMENT 02: EARLY DETECTION AND PREDICTION	PARTIALLY COMPLIANT (50%)	Priority: P0 — Critical
Current Implementation:	EarlyDetectionDashboard provides a dedicated risk view classifying centres by risk level (Critical, High, Moderate), stability scores, jitter metrics, and specific observed indicators (e.g. 5 latency spikes >210ms over 45 min). Each centre features preventive recommendations (e.g. switch to standby fiber uplink).	Verified In: Verified EarlyDetectionDashboard, risk filtering, and preventive advisory dispatch notifications.
Identified Gap:	Prediction capability is currently represented visually through static data and requires implementation. Disruption probabilities (84%, 62%) and stability scores are hardcoded in initialPredictiveCentres rather than calculated via rolling statistical analysis of incoming pings.	Impact: If an evaluator inspects whether the system actually predicts failure before it happens, they will find that numbers are pre-populated rather than computed from real-time network trends.
Actionable Recommendation:	Implement a client-side rolling buffer tracking network ping times during the exam. When standard deviation of latency exceeds a threshold, dynamically compute and display a predictive risk probability.	Next Sprint
REQUIREMENT 03: AUTOMATED INCIDENT DETECTION, CLASSIFICATION AND ESCALATION	PARTIALLY COMPLIANT (Strong Partial) (75%)	Priority: P1 — High
Current Implementation:	IncidentCenter contains an incident lifecycle tracker: Detected → Classified → Escalated → Mitigating → Resolved. Incidents are categorized (Server Failure, Network Failure, Power Failure) with severity badges, sub-second audit timelines, and root cause descriptions. Notifications are broadcast to admin and candidate channels via ResilienceContext.	Verified In: Verified IncidentCenter timeline milestones, escalation target assignment ('Both', 'Centre Supervisor'), and notification toasts.

Identified Gap:	Incident triggering is tied to the manual 'Simulate Outage' button rather than autonomous watchdog heartbeats. Escalations trigger internal React state toasts instead of external webhooks or emails.	Impact: The workflow is visually complete and demonstrable, but lacks autonomous triggering from background error thresholds.
Actionable Recommendation:	Add an autonomous heartbeat monitor in ResilienceContext that automatically triggers an incident if ping latency or packet loss crosses predefined threshold parameters.	Next Sprint
REQUIREMENT 04: BACKUP AND DISASTER RECOVERY	PARTIALLY COMPLIANT (Strong Partial) (75%)	Priority: P1 — High
Current Implementation:	RecoveryCenter showcases the 6-Phase resilience lifecycle. During an outage, the candidate's responses lock into ResponseProtectionWidget with an encrypted state hash. The exam timer freezes immediately, and upon reconnection, delta synchronization restores connectivity with zero response loss. A Controlled Resumption modal allows invigilator verification.	Verified In: Verified ResponseProtectionWidget 6-stage lifecycle, Simulate Outage / Restore Network toggle, timer freeze, and Controlled Resume approval modal.
Identified Gap:	The client sandbox stores encrypted payloads in React state and localStorage rather than a dedicated, persistent IndexedDB store with physical cryptographic key pairs. Multi-tier disaster fallback is demonstrated with mock records.	Impact: If a student closes their entire browser window during an outage (rather than just losing network), sessionStorage may reset unless saved into persistent IndexedDB.
Actionable Recommendation:	Wrap answer persistence with idb-keyval to write every selected answer directly to browser IndexedDB with a local timestamp and SHA-256 seal.	Next Sprint
REQUIREMENT 05: SECURE AND TAMPER-EVIDENT DATA STORAGE	PARTIALLY COMPLIANT (50%)	Priority: P0 — Critical
Current Implementation:	AuditTrust displays a Merkle root hash (0x7f9a842b...), candidate integrity scores (100%), and immutable audit access logs. It includes an interactive 'Simulate Tamper' button that simulates an unauthorized SQL injection attempt, immediately flags 'TAMPER_PREVENTED', and logs an access violation.	Verified In: Verified AuditTrust interface, Merkle root display, access logs table, and 'Simulate Tamper' test button.
Identified Gap:	The Merkle root is a static string in sampleAuditTrail rather than dynamically computed from the actual answers object using window.crypto.subtle. The tamper simulation injects a mock log rather than failing a real cryptographic validation.	Impact: Security auditors will recognize that the Merkle tree and WORM locking are conceptual demonstrations rather than mathematical validations over actual data.
Actionable Recommendation:	Implement an actual WebCrypto function that hashes the candidate's answers object on every save and displays the real computed SHA-256 hash in the UI.	Next Sprint
REQUIREMENT 06: INTELLIGENT IDENTIFICATION OF SUSPICIOUS PATTERNS	FULLY COMPLIANT (100%)	Priority: P2 — Medium

Current Implementation:	AIProctoringHUD runs real-time TensorFlow COCO-SSD object detection directly in the browser via webcam, detecting cell phones, multiple persons, face absence, and gaze deviation. It integrates Web Audio API to detect voice/speech noise (>65 dB) and optical flow canvas analysis for sudden physical movement. Crucially, it adheres to ethical guidelines: it flags suspicious behavior for human invigilator review rather than issuing automated disqualifications.	Verified In: Verified live COCO-SSD loading, bounding boxes on webcam, microphone audio levels, motion scores, strike counters, and routing to the invigilator review queue in LiveCandidateMonitor.
Identified Gap:	Cross-examination longitudinal pattern correlation across different test sessions uses demonstration datasets.	Impact: Exceeds standard hackathon prototypes by running real computer vision and acoustics locally without API costs.
Actionable Recommendation:	Maintain current architecture; add a simple toggle allowing evaluators without webcams to use a simulated proctoring feed if permissions are denied.	Next Sprint

REQUIREMENT 07: AUTOMATED RECONCILIATION AND VALIDATION	PARTIALLY COMPLIANT (50%)	Priority: P1 — High
Current Implementation:	ReconciliationCenter compares client-side saved response counts against cloud-received submissions. It displays mismatch statuses (Delayed Sync, Packet Drop, Exact Match). An interactive 'Run Reconciliation' button simulates a cross-system validation run, updating all records to 'Verified Reconciled' with Merkle seal confirmation.	Verified In: Verified ReconciliationCenter comparison table, discrepancy filter, and 'Run Reconciliation' engine execution.
Identified Gap:	Reconciliation operates on an in-memory mock record array rather than performing a binary or JSON delta comparison between client storage and server logs.	Impact: Demonstrates the concept and operational workflow clearly, but the matching engine is not yet connected to a live multi-node database.
Actionable Recommendation:	Connect the reconciliation button to compare the candidate's actual local answers count against the server's received answers count in ResilienceContext.	Next Sprint

REQUIREMENT 08: CANDIDATE COMMUNICATION AND REAL-TIME STATUS UPDATES	FULLY COMPLIANT (100%)	Priority: P3 — Low
Current Implementation:	LiveExam and ResponseProtectionWidget provide clear, anxiety-reducing status communication. During disruptions, candidates see an immediate status change ('Connection Lost — Responses Protected Locally'), the exam timer freezes, and an informative compensatory credit badge appears. Candidates can request assistance (rough paper, water, tech support) via StudentAssistanceModal, and receive officer broadcast messages instantly via NotificationToast.	Verified In: Verified live response protection widget, offline status transitions, help request modal, and two-way broadcast announcement toast system.
Identified Gap:	None for prototype and pilot compliance.	Impact: Superb candidate experience that directly satisfies the core requirement of transparent, reassuring communication.
Actionable Recommendation:	Maintain existing design; current implementation is presentation-ready.	Next Sprint

REQUIREMENT 09: DECISION SUPPORT FOR RESCHEDULING OR RE-CONDUCTING	PARTIALLY COMPLIANT (Strong Partial) (75%)	Priority: P1 — High
Current Implementation:	DecisionSupportCenter assists authorities in answering whether an interrupted exam can safely continue. It provides impact metrics (180 candidates affected, 100% data recovery, 1m 45s outage duration) and concrete system recommendations: 'Resume with Parity', 'Extend by 5m', 'Reschedule', or 'Re-conduct'. An official sign-off modal records administrative approvals with rationale and timestamps.	Verified In: Verified DecisionSupportCenter, recommendation cards, official sign-off modal, and administrative authorization logging.
Identified Gap:	Recommendations are based on predefined mock scenario properties rather than an automated algorithmic score assessing response loss percentages in real time.	Impact: Strong operational demonstration that proves ExamResQ understands the complex administrative dilemma of post-disruption governance.
Actionable Recommendation:	Create a simple dynamic scoring rule: if data loss = 0% and duration < 5m → recommend 'Resume with Parity'; if data loss > 10% → recommend 'Re-conduct'.	Next Sprint
REQUIREMENT 10: FAIRNESS AND CONSISTENCY DURING DISRUPTIONS	FULLY COMPLIANT (100%)	Priority: P2 — Medium
Current Implementation:	ExamResQ protects candidates from unfair penalties caused by technical faults. The system freezes candidate exam timers during outages and automatically calculates compensatory time credit (+compensatoryTimeAdded) upon recovery. Invigilators can grant +5m parity compensation with one click. Infrastructure events (network drops) are strictly segregated from cheating strikes, ensuring technical failures never result in penalties. A student grievance resolution workflow is included.	Verified In: Verified automatic timer freeze, compensatory parity calculation, manual +5m parity grant button, grievance ticket resolution, and separation of technical vs proctoring alerts.
Identified Gap:	None. The requirement is deeply embedded across both candidate and officer workflows.	Impact: Exemplary fulfillment of candidate equity and parity mandates.
Actionable Recommendation:	Highlight this feature prominently in presentations, as fairness during outages is a major judging criteria.	Next Sprint
REQUIREMENT 11: POST-EXAMINATION AUDIT TRAIL & EVIDENCE REPORTING	PARTIALLY COMPLIANT (Strong Partial) (75%)	Priority: P1 — High
Current Implementation:	Reports and AuditTrust generate a comprehensive audit trail detailing session history, timestamps, answered counts, integrity percentages, and Merkle root hashes. Upon submission in LiveExam, SubmissionReceiptModal produces an immutable digital receipt with verification tokens and integrity certificates.	Verified In: Verified Reports summary, AuditTrust event history table, and SubmissionReceiptModal verification tokens.
Identified Gap:	Reports are displayed on screen but lack a one-click 'Download Official PDF Report' button to generate exportable files for external review boards.	Impact: Judges and evaluators who expect a physical/PDF audit dossier will find that reports are currently constrained to the browser DOM.
Actionable Recommendation:	Add a client-side PDF export button (using jsPDF or the backend report generator) to export the session audit trail as a downloadable PDF document.	Next Sprint

4. End-to-End Incident Simulation Audit

A critical test of an assessment resilience platform is its ability to handle a complete end-to-end incident lifecycle without human panic or data loss. The table below audits the 15 chronological steps of an exam disruption.

Step	Expected Behavior	Current ExamResQ Implementation	Status	Required Action
1	Candidate starts exam	CandidateVerificationModal sets student station, roll, and biometrics.	WORKING	None. Seamless start.
2	Candidate answers questions	LiveExam updates active question, answers record, and SHA-256 seal.	WORKING	None. Answers persist in context.
3	Network disruption occurs	'Simulate Outage' button severs connection; offline flag engaged.	WORKING	Add autonomous heartbeat trigger.
4	Incident detected sub-second	Uplink drop flagged in 1.2s; networkStatus switches to 'disconnected'.	WORKING	None. Telemetry registers event.
5	Incident classified & prioritized	IncidentCenter records '#INC-2026-NET-402' (High Severity).	PARTIAL	Link dynamic trigger to incident log.
6	Administrator notified	NotificationToast alerts officer room; station turns amber in monitor.	WORKING	None. Alert is instant.
7	Candidate receives status update	ResponseProtectionWidget shows 'Connection Lost — Responses Protected'.	WORKING	None. Clear, non-alarming UX.
8	State checkpointed locally	Responses held in local ledger with cryptographic hash display.	PARTIAL	Store in persistent IndexedDB store.
9	Recovery initiated	'Restore Network' engages secondary route switchover.	WORKING	Add auto-retry polling reconnect.
10	State restored & synced	Protection stage moves to 'Synchronizing'; offline queue flushes to 0.	WORKING	None. Delta sync resolves.
11	Integrity validated mathematically	Status transitions to 'Response Verified: 0% Data Loss'.	PARTIAL	Recalculate dynamic Merkle root.
12	Candidate resumes with parity	Exam timer unfreezes; compensatory time (+seconds) added automatically.	WORKING	None. Fairness achieved.
13	Event logged in audit trail	CandidateTelemetryEvent '#evt-2' logged with timestamp.	WORKING	None. Appears in audit ledger.
14	Administrator reviews outcome	DecisionSupportCenter allows sign-off on parity and session extension.	WORKING	None. Full sign-off modal.
15	Audit report generated	SubmissionReceiptModal displays digital certificate upon exam submit.	PARTIAL	Add physical PDF export trigger.

5. Security, Trust & Governance Audit

Audit Dimension	Observed Implementation	Security Risk / Gap	Hardening Action
Data Integrity	SHA-256 block hashes and Merkle root displays in AuditTrust and ResponseProtectionWidget.	Hashes are computed in JavaScript state rather than signed with an asymmetric private key.	Generate Ed25519 WebCrypto keypair during student verification.
Access Control (RBAC)	OfficerLoginModal with PIN '2026'; role switching restricted in UI.	Authentication is stored in localStorage ('examresq_officer_auth'); no backend JWT verification.	Implement session expiry token in sessionStorage.

Tamper Evidence	Interactive 'Simulate Tamper' test button appends an access violation log.	Simulated log rather than rejection of a genuine memory write mutation.	Store hashed checkpoints in Object.freeze() state arrays.
Candidate Fairness	Automatic clock pause, compensatory time parity, and grievance resolution tickets.	Parity calculation is client-side; could theoretically be manipulated if unsealed.	Officer must sign off on compensatory time grants (already implemented).
AI Ethical Proctoring	TensorFlow vision & audio alerts route to human invigilator review queue.	Lighting changes can cause transient false-positive look-away frames.	Maintain the 15-frame debounce filter (already in AIProctoringHUD).

6. UX & Hackathon Demonstration Audit

ExamResQ demonstrates exceptional UX design for competitive presentations. The user interface achieves high judge-readiness through balanced whitespace, crisp typography, and disciplined dummy data (strictly limited to 3 items per entity). Screens communicate their story within 3 to 5 seconds without developer jargon. The 2-minute pitch demonstration flow is immediately compelling: 1. *Start Exam* → 2. *Pull plug / Simulate Outage* → 3. *Show timer freeze & protected local ledger* → 4. *Restore Network* → 5. *Show zero loss & audit receipt*.

7. Missing Features — Actionable Engineering Backlog

P	Feature Name	Target Module	Why Needed for Compliance	Actionable Implementation Steps
P0	Dynamic Latency Risk Scoring	EarlyDetectionDashboard	Req 2: Prediction must not be hardcoded static data.	Record rolling ping latency array in ResilienceContext; calculate standard deviation; compute risk probability = min(95, std_dev * 1.8).
P0	Live WebCrypto SHA-256 Hashing	ResponseProtectionWidget, AuditTrust	Req 5: Mathematical proof of answer tamper-evidence.	Call window.crypto.subtle.digest('SHA-256', JSON.stringify(answers)) on every option click; display real hex string.
P1	Persistent IndexedDB Storage	ResilienceContext, LiveExam	Req 4: Protect against browser/tab crash during outage.	Use idb-keyval to write answers to indexedDB store 'examresq_answers' on every save; restore on mount.
P1	Client-to-Server Reconciliation Diff	ReconciliationCenter	Req 7: Real validation of local vs remote deltas.	Compare Object.keys(answers).length against mock cloud submitted count; show true reconciliation diff.
P1	PDF Audit Dossier Export	Reports, AuditTrust	Req 11: Exportable proof for government audit boards.	Add 'Download Official Audit Dossier' button in Reports triggering print-friendly HTML/PDF export.
P2	Autonomous Outage Watchdog	ResilienceContext	Req 3: Incidents should trigger from background watchdog.	Add an optional 'Autonomous Watchdog Mode' toggle that simulates network dropout after 45 seconds of exam activity.
P3	Webcam Permission Fallback Simulator	AIProctoringHUD	Req 6: Allow presentation when webcam is disabled/blocked.	If navigator.mediaDevices.getUserMedia fails, fallback to looping sample proctoring canvas with demo detection boxes.

8. Implementation Roadmap

PHASE 1: Critical Compliance (Sprint 1)	• Implement real WebCrypto SHA-256 hashing for student answers. • Connect dynamic variance calculation to Early Detection Dashboard risk probabilities. • Store answers in IndexedDB alongside React context.
PHASE 2: Strong Demonstration (Sprint 2)	• Connect Reconciliation Center to actual candidate answers delta. • Add one-click official PDF report export in Reports & AuditTrust. • Provide demo fallback video loop for AI Proctoring if webcam blocked.

PHASE 3: Production Hardening (Future Pilot)	• Multi-region WebSocket signaling cluster for nationwide peer-to-peer mesh. • Hardware Security Module (HSM) digital signatures for submission receipts. • Formal institutional API integrations with NTA / State Board enrollment databases.
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9. Final Audit Verdict

VERDICT: ALMOST READY (75.0% COMPLIANT)

ExamResQ demonstrates a world-class concept, stellar visual execution, and genuinely functional frontend innovations (real-time TensorFlow proctoring, Web Audio decibel meters, BroadcastChannel mesh synchronization, and active timer parity). It is not yet graded 'FULLY READY' because 8 of the 11 requirements currently rely on mock datasets or simulated client state for backend operations (risk prediction scores, Merkle root verification, and reconciliation deltas). Executing the 2 P0 items (Dynamic Risk Scoring and Real WebCrypto Hashing) will instantly elevate ExamResQ to a near-flawless, unassailable hackathon-winning submission.